

Lab 13: Advanced NLP & Vector Search

Project: Cybersecurity Knowledge-Based QnA Bot

1. Project Overview

This project involves the development of a semantic Question-and-Answer (QnA) system focused on Cybersecurity. Unlike traditional keyword-based search engines, this bot utilizes **Large Language Models (LLMs)** and **Vector Similarity Search** to understand the intent and context behind a user's query.

The system is designed to act as a security expert, providing immediate advice on topics such as phishing, spam detection, and digital hygiene.

2. Architecture & Technologies

- **Sentence Transformers (MiniLM):** A pre-trained model from Hugging Face used to convert text into high-dimensional numerical vectors (embeddings).
- **FAISS (Facebook AI Similarity Search):** A library for efficient similarity search and clustering of dense vectors.
- **Flask:** A micro web framework for Python used to deploy the model as a user-friendly web application.
- **Pandas:** Used for data management and handling the knowledge base.

3. Implementation Workflow

3.1. Knowledge Base Construction

The foundation of the bot is a curated dataset of questions and answers. Each entry represents a specific cybersecurity scenario. This data is preprocessed to remove special characters and normalized to lowercase to ensure consistency during the embedding process.

3.2. Semantic Embedding (Vectorization)

Using the '*paraphrase-MiniLM-L6-v2*' model, each question in the knowledge base is transformed into a vector. These vectors capture the "meaning" of the sentence. For example, the phrases "How to stop spam" and "Ways to block junk mail" would result in vectors that are mathematically close to each other.

3.3. FAISS Indexing

The generated vectors are stored in a FAISS index. This specialized database structure allows the system to perform **Nearest Neighbor Search**. When a user asks a question, the system finds the closest mathematical match in the index in milliseconds.

3.4. Flask Integration & Real-time Retrieval

The web interface accepts user input and triggers the backend pipeline. The user's query is embedded on-the-fly, compared against the FAISS index, and the corresponding answer from the knowledge base is retrieved and displayed on the UI.

4. Key Features

- **Contextual Understanding:** The bot can answer queries even if the user does not use exact keywords found in the database.
- **Scalability:** Thanks to FAISS, the knowledge base can be scaled to thousands of entries without a significant drop in performance.

- **User-Friendly Interface:** A clean, responsive HTML/CSS frontend for seamless user interaction.

5. Conclusion

By combining Hugging Face's state-of-the-art embeddings with FAISS's efficient retrieval capabilities, this project demonstrates the modern approach to building intelligent QnA systems. This workflow is directly applicable to enterprise-level solutions such as customer support bots and internal document search engines.